

FBISE Chemistry SSC-II | Second Annual Examination 2025

Complete worked solutions: Sections A, B and C

Prepared by Fakhar STEM Sphere

The paper contains 12 MCQs (12 marks), 11 compulsory short-question parts with a choice in each (33 marks), and four long questions with a choice in each (20 marks). Both sides of every OR choice are answered below. The paper uses the conventional school-level account of these processes; equations are balanced where appropriate.

Section A — MCQs (1 mark each)

1. Bromine water test indicates the presence of all EXCEPT — B, CH₃-CH₃ (ethane).

The orange/brown colour of bromine water disappears rapidly when bromine adds across a C=C or C≡C bond. A, H₂C=CH₂ (ethene), C, HC≡CH (ethyne), and D, CH₃-CH=CH₂ (propene), are unsaturated and give the test. B is saturated: ethane has only single C-C and C-H bonds and does not decolourise bromine water under normal test conditions. A saturated alkane can undergo slow substitution under UV light, which is a different test condition.

2. Arrhenius base — C, NaOH.

NaOH → Na⁺ + OH⁻ in water, so it directly increases aqueous hydroxide ions. A, HCl, produces H⁺/H₃O⁺ and is an acid. B, NH₃, makes OH⁻ by reacting with water and is a Brønsted-Lowry base; in the strict Arrhenius definition used here it is not the hydroxide-containing compound. D, NH₄Cl, is a salt whose NH₄⁺ can acidify water.

3. Straight-chain hydrocarbon — A, C₄H₁₀ (the intended choice).

B, C₆H₆, is an aromatic ring, not a straight chain. C, CH₃CH(CH₃)CH₃, is explicitly branched 2-methylpropane. D, C₆H₁₂O₆, contains oxygen, so it is not a hydrocarbon. A can be written as straight-chain n-butane, CH₃-CH₂-CH₂-CH₃. Strictly, C₄H₁₀ alone is ambiguous because branched isobutane has the same molecular formula; the paper evidently intends its unbranched isomer.

4. Brønsted-Lowry base — C, NH₃.

NH₃ accepts a proton: NH₃ + H₂O ⇌ NH₄⁺ + OH⁻. A, H₂SO₄, donates protons. D, NaCl, is a neutral salt in this context. B, KOH, supplies OH⁻, which does accept a proton; therefore KOH solution is also basic and can be described as a source of a Brønsted-Lowry base. The question is imperfectly exclusive: C is the intended answer because NH₃ illustrates proton acceptance directly, whereas KOH is usually selected as the Arrhenius example.

5. Bond joining amino acids in proteins — C, peptide bonds.

A peptide bond, -CO-NH-, joins the carboxyl group of one amino acid to the amino group of another. A, hydrogen bonds, help stabilize protein shapes but do not form the primary amino acid chain. B, ionic bonds, may occur between charged side chains, not the backbone link. D, glycosidic bonds, join sugar units in carbohydrates.

6. Pollutant primarily responsible for acid rain — D, SO₂.

Sulfur dioxide oxidises in air and forms sulfuric acid in rain: 2SO₂ + O₂ → 2SO₃; SO₃ + H₂O → H₂SO₄. Nitrogen oxides also contribute, but none is among the choices. A, CO, does not primarily form acid rain. B, N₂, and C, O₂, are ordinary atmospheric gases, not the identified acid-rain pollutants.

7. Impurity causing temporary hardness — D, Ca(HCO₃)₂.

Calcium hydrogencarbonate decomposes on boiling to insoluble CaCO₃, so this hardness is temporary. A, CaSO₄, and C, MgCl₂, cause permanent hardness. B, NaCl, does not cause the characteristic Ca²⁺/Mg²⁺ hardness.

8. Industrial manufacture of sodium carbonate — C, Solvay process.

The Solvay process produces soda ash, Na_2CO_3 , from brine and limestone with recycled ammonia. A, Contact process, makes H_2SO_4 . B, Haber process, makes NH_3 . D, Ostwald process, makes HNO_3 .

9. Functional group of carboxylic acids — D, $-\text{COOH}$.

The carboxyl group consists of carbonyl and hydroxyl on the same carbon. A, $-\text{CHO}$, is the aldehyde group. B, $-\text{OH}$, is the alcohol hydroxyl group when attached to a carbon chain. C, $-\text{NH}_2$, is the amino group.

10. Condition favouring the forward exothermic reaction — B, low temperature.

Heat is a product of an exothermic forward reaction. Cooling favours the heat-producing direction by Le Châtelier's principle. A, high temperature, favours the endothermic reverse direction. C, high pressure, cannot be judged without gas mole counts. D, increased concentration, cannot be judged without saying which reactant or product is increased.

11. Solution turning blue litmus red — B, vinegar.

Vinegar contains ethanoic (acetic) acid, so it turns blue litmus red. A, soap, C, baking soda solution, and D, typical detergent solutions are alkaline and normally turn red litmus blue; detergent formulations can vary, but vinegar is unambiguously acidic.

12. Substance decolourising pink/purple KMnO_4 and being oxidised — B, ethene.

The $\text{C}=\text{C}$ double bond in ethene is oxidised by dilute, cold alkaline potassium permanganate (Baeyer test); the permanganate colour disappears, and ethane-1,2-diol is formed. A, ethane, C, methane, and D, butane, are saturated alkanes and do not give the ordinary unsaturation test. The wording calls KMnO_4 "pink"; dilute permanganate is commonly described as pink or purple.

Section B — Q2: all 11 parts and all OR alternatives (3 marks each)

(i) Law of mass action; equilibrium expression

At a fixed temperature, the rate of an elementary reaction is proportional to the product of the concentrations of its reactants, each raised to the power of its stoichiometric coefficient. For $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$, forward rate $k_f[\text{A}]^a[\text{B}]^b$ and reverse rate $k_r[\text{C}]^c[\text{D}]^d$. At equilibrium these rates are equal; hence $K_c = [\text{C}]^c[\text{D}]^d / ([\text{A}]^a[\text{B}]^b)$. Example: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, so $K_c = [\text{HI}]^2 / ([\text{H}_2][\text{I}_2])$. Concentrations in the expression are equilibrium concentrations.

(i) OR — self-ionisation of water and K_w

$2\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$, often shortened to $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$. Thus $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 25°C . In pure water each water molecule that ionises yields one hydronium and one hydroxide ion, so $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1.0 \times 10^{-7} \text{ mol dm}^{-3}$. Their product is $10^{-7} \times 10^{-7} = 10^{-14}$. Water molecules are omitted from K_w because liquid water has effectively constant activity.

(ii) Hydrogenation of ethene; catalyst

$\text{CH}_2=\text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3-\text{CH}_3$ (ethane), in the presence of finely divided nickel (Ni) catalyst and heat; platinum or palladium also catalyse the reaction. Hydrogen adds across the double bond, converting an alkene to an alkane.

(ii) OR — survival of aquatic life in extreme cold

Water has maximum density near 4°C . As a lake cools below 4°C , the colder water stays on top; ice is less dense than liquid water and floats. The surface ice layer insulates the water beneath, which can remain liquid near 4°C , allowing fish and other aquatic organisms to survive. Water's high heat capacity also slows temperature changes.

(iii) Ethanol and ethanal functional groups

Ethanol is $\text{CH}_3\text{-CH}_2\text{-OH}$; its functional group is hydroxyl ($-\text{OH}$), characteristic of an alcohol. Ethanal is $\text{CH}_3\text{-C(=O)-H}$, also written $\text{CH}_3\text{-CHO}$; its functional group is aldehyde ($-\text{CHO}$), a terminal carbonyl group. In ethanal the $-\text{OH}$ group is absent.

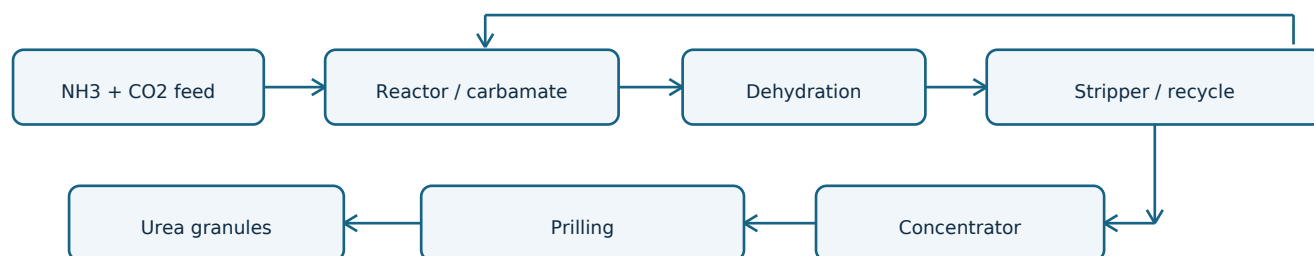
(iii) OR — how SO_x and NO_x cause acid rain

Sulfur dioxide from burning sulfur-containing fuels is oxidised: $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$; $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$. Nitrogen monoxide from high-temperature combustion is oxidised: $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$; $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$. These acids dissolve in cloud/rain water and lower its pH. SO_2 can also first form sulfurous acid, $\text{SO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{SO}_3$, before further oxidation.

(iv) Two ways to remove permanent hardness

1. Washing soda (Na_2CO_3) precipitates calcium and magnesium ions: $\text{CaSO}_4(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) \downarrow + \text{Na}_2\text{SO}_4(\text{aq})$; similarly $\text{MgCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{MgCO}_3 \downarrow + 2\text{NaCl}$. Remove the precipitate by filtration. 2. Ion-exchange resin or zeolite replaces $\text{Ca}^{2+}/\text{Mg}^{2+}$ with Na^+ : $2\text{R-Na} + \text{Ca}^{2+} \rightarrow \text{R}_2\text{-Ca} + 2\text{Na}^+$ (the same applies to Mg^{2+}). A chemical equation is supplied for each method, although the question requests one.

(iv) OR — labelled flow sheet for urea manufacture



NH_3 feed + CO_2 feed $\rightarrow$ high-pressure reactor (ammonium carbamate formation, $2\text{NH}_3 + \text{CO}_2 \rightleftharpoons \text{NH}_2\text{COONH}_4$) $\rightarrow$ dehydration ($\text{NH}_2\text{COONH}_4 \rightleftharpoons \text{NH}_2\text{CONH}_2 + \text{H}_2\text{O}$) $\rightarrow$ separator/stripper (recover unreacted NH_3 and CO_2) $\rightarrow$ evaporation/concentration $\rightarrow$ prilling or granulation $\rightarrow$ solid urea, NH_2CONH_2 . Recovered NH_3 and CO_2 are recycled to the reactor. Overall: $2\text{NH}_3 + \text{CO}_2 \rightleftharpoons \text{NH}_2\text{CONH}_2 + \text{H}_2\text{O}$. Label both feedstocks, the intermediate, recycle stream and final product in the diagram.

(v) Classify solutions by $[\text{H}^+]$

At 25°C , neutral water has $[\text{H}^+] = 1 \times 10^{-7} \text{ M}$. A: $1 \times 10^{-5} \text{ M}$ is greater, so acidic (pH 5). B: $1 \times 10^{-7} \text{ M}$ is neutral (pH 7). C: $1 \times 10^{-11} \text{ M}$ is smaller, so basic (pH 11). Since $\text{pH} = -\log_{10}[\text{H}^+]$, each tenfold reduction raises pH by one unit.

(v) OR — electrorefining of copper

Put impure copper at the positive anode, a thin sheet of pure copper at the negative cathode, and acidified aqueous CuSO_4 as electrolyte. On passing current, anode copper dissolves, $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$, while copper ions plate onto the cathode, $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$. Insoluble noble-metal impurities fall as anode mud; some reactive impurities stay dissolved. The cathode grows into high-purity copper.

(vi) Macroscopic signs of forward and reverse reactions

At first the concentrations/observable intensities of reactants decrease and those of products increase as the forward reaction dominates. As products build up, the reverse reaction also occurs: products convert back to reactants. At dynamic equilibrium, colour, pressure and measured concentrations remain constant with time although both reactions continue at equal rates. Example: in a closed $\text{NO}_2/\text{N}_2\text{O}_4$ system, brown colour becomes steady at equilibrium; steady colour does not mean the reactions stop.

(vi) OR — structures and relative reactivity of ethane and ethene

Ethane: $\text{CH}_3\text{-CH}_3$, a saturated alkane with a single C-C bond. Ethene: $\text{CH}_2=\text{CH}_2$, an unsaturated alkene with a C=C double bond. Ethene is more reactive in ordinary addition reactions because the π part of the double bond is easier to break than a σ bond. It adds bromine, for example $\text{CH}_2=\text{CH}_2 + \text{Br}_2 \rightarrow \text{BrCH}_2\text{-CH}_2\text{Br}$; ethane does not react this way without special conditions.

(vii) Three air pollutants and their sources

CO: incomplete combustion in vehicle engines and poorly ventilated stoves. SO_2 : burning sulfur-containing coal/oil in power stations or smelting sulfide ores. NO_x (NO and NO_2): high-temperature vehicle engines and power plants where atmospheric N_2 and O_2 react. State both the pollutant and its source for each mark.

(vii) OR — three uses of urea

1. Nitrogen fertiliser for crop growth (urea supplies nitrogen). 2. Raw material for urea-formaldehyde resins used in wood adhesives/particle board. 3. Raw material or component in certain plastics, coatings and other chemical products. Urea is also used in diesel exhaust fluid to help reduce NO_x emissions.

(viii) Three petroleum fractions with uses

Refinery gas/LPG: cooking and heating fuel. Gasoline/petrol: fuel for spark-ignition cars and motorcycles. Kerosene: jet fuel or heating/lighting fuel. Other acceptable examples include diesel for heavy vehicles, lubricating oil for machinery, and bitumen for roads.

(viii) OR — three conditions for chemical equilibrium

1. A reversible reaction must be possible in both directions. 2. The system must be closed, so reactants and products do not escape. 3. At a fixed temperature, forward and reverse rates become equal; concentrations then remain constant (although not necessarily equal). A catalyst may speed the approach to equilibrium but is not required for equilibrium.

(ix) Raw materials of Solvay process

Concentrated sodium chloride solution (brine, NaCl), limestone (CaCO_3), and ammonia (NH_3). Water is also present. Ammonia is largely recycled, while brine supplies Na^+ and limestone supplies CO_2 . Listing brine, limestone and ammonia meets the requested three materials.

(ix) OR — three waterborne diseases and causes

Cholera: *Vibrio cholerae* bacteria in contaminated drinking water. Typhoid fever: *Salmonella Typhi* bacteria spread by faecally contaminated water. Hepatitis A: hepatitis A virus spread by contaminated water (hepatitis E is another waterborne viral example). Poor sanitation and sewage contamination enable transmission; each disease should be paired with its causative organism.

(x) Three effects of household/industrial wastewater

1. Organic sewage increases biological oxygen demand, lowers dissolved O_2 and can kill fish. 2. Excess nitrates/phosphates trigger algal blooms (eutrophication), followed by oxygen depletion. 3. Toxic industrial substances such as heavy metals poison aquatic life and can accumulate in food chains. Pathogens can also spread waterborne illness.

(x) OR — three alkyl radicals from butane using structures

Removing one H from n-butane gives butan-1-yl, $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\cdot$, and butan-2-yl, $\text{CH}_3\text{-CH}_2\text{-CH}\cdot\text{-CH}_3$. If "butane" includes its branched isomer 2-methylpropane, removing H from a terminal CH_3 gives 2-methylpropan-1-yl, $(\text{CH}_3)_2\text{CH-CH}_2\cdot$; removing its central H gives 2-methylpropan-2-yl, $(\text{CH}_3)_3\text{C}\cdot$. Any three of these four $\text{C}_4\text{H}_9\cdot$ structural radicals answer the intended question. If the instruction means only straight-chain n-butane, exactly two distinct positional radicals exist; the third necessarily comes from its branched isomer. The dot marks the unpaired electron/free valence.

(xi) Ethyne from 1,2-dichloroethane

Successive dehydrohalogenation removes two molecules of HCl. First, $\text{ClCH}_2\text{-CH}_2\text{Cl} + \text{KOH(alc), heat} \rightarrow \text{CH}_2=\text{CHCl} + \text{KCl} + \text{H}_2\text{O}$ (chloroethene/vinyl chloride). Then $\text{CH}_2=\text{CHCl} + \text{NaNH}_2$ in liquid $\text{NH}_3 \rightarrow \text{HC}\equiv\text{CH} + \text{NaCl} + \text{NH}_3$. Strong base is preferred for the second elimination because a vinyl halide is less easily dehydrohalogenated than an ordinary alkyl halide. At introductory level the net equation is often written $\text{ClCH}_2\text{-CH}_2\text{Cl} + 2\text{KOH(alc)} \rightarrow \text{HC}\equiv\text{CH} + 2\text{KCl} + 2\text{H}_2\text{O}$ under vigorous conditions.

(xi) OR — rate constant for $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$

Assuming the rate law intended by the question is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$, rearrange to $k = \text{rate}/([\text{NO}]^2[\text{O}_2])$. Rate has units $\text{mol dm}^{-3} \text{ s}^{-1}$; $[\text{NO}]^2[\text{O}_2]$ has units $(\text{mol dm}^{-3})^3$. Hence $[k] = (\text{mol dm}^{-3} \text{ s}^{-1})/(\text{mol}^3 \text{ dm}^{-9}) = \text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ (equivalently $\text{L}^2 \text{ mol}^{-2} \text{ s}^{-1}$). The balanced equation alone does not generally determine reaction orders; these exponents require an experimentally established rate law or an explicit elementary-step assumption. State this assumption for full scientific accuracy.

Section C — Q3-Q6 with every OR alternative**Q3. Lewis acids and bases versus Brønsted-Lowry acids and bases (2 + 2 + 2 marks)**

A Lewis acid accepts an electron pair, whereas a Lewis base donates an electron pair. A Brønsted-Lowry acid donates H^+ ; a Brønsted-Lowry base accepts H^+ . Thus Lewis theory describes electron-pair transfer and includes reactions with no proton, whereas Brønsted-Lowry theory specifically describes proton transfer. Lewis example: $\text{BF}_3 + \text{:NH}_3 \rightarrow \text{F}_3\text{B}\leftarrow\text{NH}_3$; electron-deficient BF_3 is the acid and NH_3 the base. Brønsted example: $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$; HCl donates H^+ (acid), H_2O accepts it (base). $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ also illustrates NH_3 as a proton-accepting base.

Q3 OR. Typical amino acid and tripeptide (2 + 3 + 1 marks)

General α -amino acid: $\text{H}_2\text{N-CH(R)-COOH}$, with an amino group ($-\text{NH}_2$), carboxyl group ($-\text{COOH}$), H and variable side chain R attached to the central carbon. In aqueous solution the zwitterion $\text{H}_3\text{N}^+\text{-CH(R)-COO}^-$ is common. Condensation of three amino acids $\text{H}_2\text{N-CH(R}_1\text{)-COOH}$, $\text{H}_2\text{N-CH(R}_2\text{)-COOH}$ and $\text{H}_2\text{N-CH(R}_3\text{)-COOH}$ produces $\text{H}_2\text{N-CH(R}_1\text{)-CO-NH-CH(R}_2\text{)-CO-NH-CH(R}_3\text{)-COOH} + 2\text{H}_2\text{O}$. The two $-\text{CO-NH}-$ links are peptide bonds. Three amino acid units form a tripeptide; two water molecules are eliminated. A concrete example is glycylalanylglycine ($\text{R}_1=\text{H}$, $\text{R}_2=\text{CH}_3$, $\text{R}_3=\text{H}$).

Q4. Monosaccharides and disaccharides: physical properties and examples (3 + 3 marks)

Monosaccharides are single sugar units and cannot be hydrolysed into simpler carbohydrates; examples are glucose and fructose (both $\text{C}_6\text{H}_{12}\text{O}_6$). They are usually crystalline, sweet and soluble in water because of multiple $-\text{OH}$ groups. Disaccharides consist of two monosaccharide units joined by a glycosidic bond and can be hydrolysed; examples are sucrose and maltose (both $\text{C}_{12}\text{H}_{22}\text{O}_{11}$). They are generally crystalline, sweet and water-soluble too; there is overlap, so sweetness/solubility alone cannot reliably identify the class. Disaccharides have larger molecules and typically different melting/thermal-decomposition behaviour; the decisive comparison is one unit versus two units and ability to hydrolyse. Example: $\text{sucrose} + \text{H}_2\text{O} \rightarrow \text{glucose} + \text{fructose}$, catalysed by acid or sucrase.

Q4 OR. Preparing ethene by two methods (3 + 3 marks)

(a) Dehydration of ethanol: $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$, heated with concentrated H_2SO_4 around 170°C (443 K), or ethanol vapour over heated Al_2O_3 . Concentrated acid acts as dehydrating agent; 140°C can favour ether instead, so state the higher temperature. (b) Dehydrohalogenation of an ethyl halide: $\text{CH}_3\text{CH}_2\text{Br} + \text{KOH(ethanolic), heat} \rightarrow \text{CH}_2=\text{CH}_2 + \text{KBr} + \text{H}_2\text{O}$. Alcoholic KOH removes a hydrogen from one carbon and Br from its neighbour (elimination); aqueous KOH tends to give substitution to ethanol instead. $\text{CH}_3\text{CH}_2\text{Cl}$ may be used with KCl as by-product.

Q5. Petroleum and its formation (4 marks)

Petroleum/crude oil is a naturally occurring complex mixture of hydrocarbons, mainly alkanes and cycloalkanes, with smaller amounts of other compounds; it is found in porous underground rock, often with natural gas and water. Ancient marine organisms died and accumulated beneath sediments. Over millions of years, burial in low-oxygen conditions, heat and pressure converted their organic remains to hydrocarbons. These hydrocarbons migrated through permeable rock and were trapped beneath an impermeable cap rock, forming an oil reservoir. Crude oil is separated into useful fractions by fractional distillation according to boiling ranges.

Q5 OR. Troposphere versus stratosphere (2 + 2 marks)

Troposphere: lowest atmospheric layer, from the surface to roughly 8–18 km depending on latitude. It contains most atmospheric air and nearly all water vapour, clouds and weather; temperature generally falls with altitude. It provides breathable air, supports the water cycle and determines surface weather, while pollutants here can damage health and ecosystems. Stratosphere: layer above the troposphere, extending to about 50 km. It contains the ozone layer, with comparatively little water vapour; temperature generally increases upward as ozone absorbs solar UV. Its ozone screens biologically harmful UV and protects life. Both layers are predominantly N_2 and O_2 ; the distinction is their water vapour/ozone distribution and environmental roles.

Q6. Removal of temporary hardness, with equations (2 + 2 marks)

Method 1, boiling: $Ca(HCO_3)_2(aq) \rightarrow CaCO_3(s) \downarrow + CO_2(g) \uparrow + H_2O(l)$. Magnesium bicarbonate can give $Mg(HCO_3)_2 \rightarrow MgCO_3 + CO_2 + H_2O$, followed by $MgCO_3 + H_2O \rightarrow Mg(OH)_2 \downarrow + CO_2$. Insoluble deposits are separated by filtration. Method 2, Clark's/lime method: add calculated $Ca(OH)_2$ to precipitate bicarbonate hardness: $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 \downarrow + 2H_2O$; $Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow Mg(OH)_2 \downarrow + 2CaCO_3 \downarrow + 2H_2O$. Remove the precipitates by settling/filtration; avoid adding an unnecessary excess of lime.

Q6 OR. Stratospheric ozone formation, degradation, and UV role (3 + 1 marks)

Formation: solar UV splits oxygen, $O_2 + h\nu \rightarrow 2O\cdot$; an oxygen atom combines with O_2 in the presence of a third body M, $O\cdot + O_2 + M \rightarrow O_3 + M$. Natural loss occurs when O_3 absorbs UV, $O_3 + h\nu \rightarrow O_2 + O\cdot$, and when $O\cdot + O_3 \rightarrow 2O_2$. Human-released CFCs can increase destruction: UV frees $Cl\cdot$ from a CFC; $Cl\cdot + O_3 \rightarrow ClO\cdot + O_2$; $ClO\cdot + O\cdot \rightarrow Cl\cdot + O_2$. Net: $O_3 + O\cdot \rightarrow 2O_2$; chlorine is regenerated and can attack further ozone. The ozone layer absorbs nearly all harmful UV-C and much UV-B, reducing DNA damage, skin cancer/cataract risk and harm to crops and marine organisms. Some UV-A and UV-B still reach the ground.

Exam-use notes

Write the balanced equation, name the reagent/catalyst or condition, and state the key observation or conclusion where applicable. Draw structural formulas clearly with the correct number of bonds. For Q2(iv) OR, draw boxes and arrows for the flow sheet using the labelled sequence above. For Q2(x) OR and Q2(xi) OR, the source questions contain genuine scientific ambiguity; the notes above identify the intended school answer and its necessary qualification.